

Group Assignment - Micromouse

MTRN3100 - UNSW School of Mechanical and Manufacturing Engineering

Changelog

Learning Outcomes

- Apply relevant theoretical knowledge about mobile robots, including locomotion, perception, and localisation, utilising onboard sensors, navigation, and path planning, for complex problem-solving.
- Apply general computer vision techniques for feature/object detection and tracking, for complex problem-solving.
- Demonstrate hands-on skills in mechatronics design, fabrication, and implementation by completing practical activities.
- Collaborate effectively within a team via participation in a problem-solving competition.

1 Objective

The objective of the micromouse challenge is to build a robot that navigates through a maze in the shortest time. Each team's run will be recorded and ranked to decide an overall winner.

1.1 Robot Specification

- The robot must be powered by the provided battery pack, with the appropriate fuse.
- All software processing must be done with the onboard Arduino nano microcontroller unless otherwise specified (computer vision for path generation).
- The robot must fit within a cylinder 150 mm wide and 150 mm tall.
- The robot must not use permanent fastening techniques e.g. epoxy and superglue.

Sufficient robot components are supplied; however, you are allowed to include your own components within reason. If you are unsure if additional components are allowed, please contact the course staff. Replacement parts will not be provided unless you can prove your component was damaged through no fault of your own.

1.2 Maze representation

You are free to represent the maze however you choose. However, it must be autonomously generated from your computer vision solution. A source and destination location will be provided. The complete maze comprises cells in a 9 x 9 configuration. Each cell is 180mm x 180mm, and the grid is enclosed with a wall height of 150mm (Figure. 1).

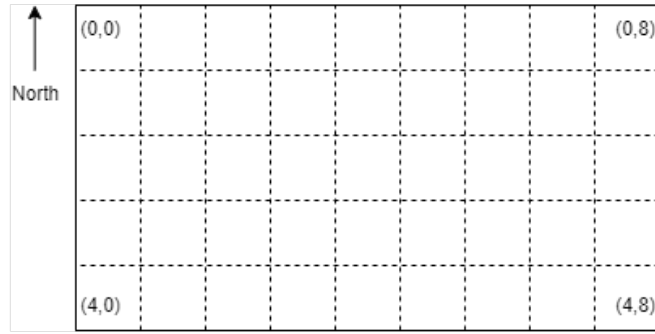


Figure 1: Representation of the maze grid with coordinates.

The start position is given in the configuration (row, column, direction), which indicates the robot's initial row and column (zero-indexed) as well as direction (N, S, E, W). If the start position is (1,6,N) the robot will be starting in the 2nd row, 7th column, pointing north. The goal position is given in the format (row, column), which indicates the target row and column.

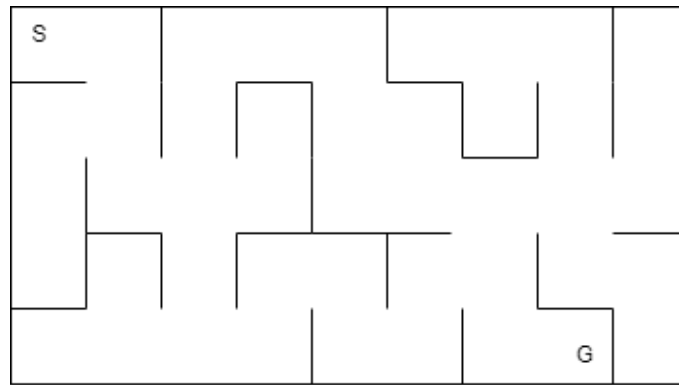


Figure 2: Example maze layout for the final competition. S represents the start location at cell (0,0). G represents the goal location at cell (4,7).

2 Barebones movement (Week 4) - 3% Total

The first milestone involves demonstrating autonomous straight-line locomotion and accurate in-place rotation on any platform. A complete robot is not required; you simply need two actuated wheels on any movable platform to achieve locomotion. The two tasks below must be completed in the same program with no restarting between them. Manual intervention at any point after starting the robot will result in 0 marks for that run. Each team receives a maximum of 2 runs, with your best attempt used for marking. This assessment will be marked during your week 4 lab time, and is worth 3% of your final course mark.

First, you must drive a distance of 200mm within a reasonable amount of time. A 1m wide start line will be taped on the ground, with a 1m wide line taped 200mm from it defining the goal zone. Place your robot along the start line with both wheels touching the tape, then turn it on and drive to the goal zone before coming to rest. You achieve 1 mark if all parts of your robot are within ± 75 mm of the goal line when it comes to rest.

Second, without restarting, your robot must rotate in place. The heading at the start of this task defines the reference heading. The robot must perform four consecutive 90-degree counter-clockwise (left) turns, then four consecutive 90-degree clockwise (right) turns. You achieve 1 mark if the robot is at rest within ± 45 degrees of the reference heading after the four left turns, and a further 1 mark if it is again within ± 45 degrees after the four right turns. The testing time will be capped at one minute per task; exceeding this results in 0 marks for that task. Marks and measurements are at the discretion of the demonstrator.

3 Simple driving (Week 8) - 7% Total

The simple driving task involves 4 different tasks. You may compile and upload code between trials; however, no significant changes will be allowed. A significant change is one requiring more than 5 minutes to implement. This assessment will be marked during your week 8 lab time, and is worth 7% of your final course mark.

3.1 Straight-line tracking - 2%

In this task, you must demonstrate that your robot can drive in a straight line over an extended distance without significant lateral drift. A start line will be taped on the ground and a finish line taped 1m from it, with a centreline marking the intended path of travel. Place your robot on the start line, centred on the centre line, then turn it on and drive 1m forward until it comes to rest past the finish line. You achieve 2 marks if the robot reaches the finish line with its centre having remained within $\pm 100\text{mm}$ of the centre line throughout the run, or 1 mark if its centre strays beyond $\pm 100\text{mm}$ but remains within $\pm 200\text{mm}$ throughout. If your robot requires manual intervention, strays beyond $\pm 200\text{mm}$, or does not reach the finish line within 30 seconds, the run will be terminated. Each team receives a maximum of 2 runs for this task, with your best attempt used for marking. Marks and measurements are at the discretion of the demonstrator.

Tip: Don't try to solve this task with open-loop control; use the IMU.

3.2 Driving and stopping - 2%

In this task, you will need to use the front lidar and a distance-based controller to accurately move to a fixed distance from a wall. This task consists of three sequential challenges that must be completed in the same program (no restarting between challenges). 1 mark will be awarded for each completion of each challenge. If your robot requires manual intervention at any stage or does not reach the desired state within 10 seconds, the run will be terminated.

First, the robot will be placed such that the front of the robot is 200mm away from a static wall. It must drive forward until the front of the robot is 100mm $\pm 5\text{mm}$ from the wall and at rest. Second, the demonstrator will slowly slide the wall back 100mm, and the robot must follow it, trying to maintain a distance of 100mm before eventually coming to rest 100mm $\pm 5\text{mm}$ from the wall. Finally, the demonstrator will slowly slide the wall forward 100mm, and the robot must follow it, trying to maintain a distance of 100mm before eventually coming to rest 100mm $\pm 5\text{mm}$ from the wall. The demonstrator will not deliberately move the wall into the robot; however, you must achieve the desired setpoint within 10 seconds after the wall is moved into place. Each team receives a maximum of 2 runs for this task, with your best attempt used for marking. Marks and distance measurements are at the discretion of the demonstrator.

3.3 Turning - 2%

In this task, you must be able to demonstrate your ability to accurately turn, either with lidars and/or an IMU. You have the choice of where this test takes place, either on the floor with no walls or within a single cell with walls on all 4 sides. This task consists of two sequential challenges that must be completed in the same program (no restarting between challenges). 1 mark will be awarded for each completion of each challenge. If your robot requires manual intervention at any stage or does not reach the desired state within 10 seconds, the run will be terminated.

First, your robot must perform a 90-degree clockwise turn when you turn it on. You will receive 1 mark if, after 10 seconds, the robot is at rest and within ± 5 degrees of the desired rotation. Second, the demonstrator will raise the robot $>50\text{mm}$ above the ground and turn it up to 120 degrees in either direction. The demonstrator will lower the robot onto the floor, where the robot must rotate back to the setpoint. You will receive 1 mark if, after 10 seconds, the robot is at rest and within ± 5 degrees of the desired rotation. Each team receives a maximum of 2 runs for this task, with your best attempt used for marking. Marks and angle measurements are at the discretion of the demonstrator.

3.4 Chaining Movements - 1%

You will be provided a string of 8 movements that your robot must be able to execute in a standard maze environment. For the maze shown in Figure. 2, the robot may start at (0,0,S) and be provided the string "lfrfflfr". This string would be the command sequence "left, forward, right, forward, forward, left, forward, right". In which forward indicates moving one cell, Left is a turn 90 degrees counter-clockwise, and Right is a turn 90 degrees clockwise.

Achieving at least 4 actions is worth 0.5 marks, while all 8 actions are worth the full 1 mark. A collision will terminate your run. Each team receives a maximum of 2 runs for this task, with your best attempt used for marking. The command string will be provided on the day of marking.

3.5 Submission

Marking will occur during your Week 8 Lab. Teams will be called to the front to be marked. It is expected that you turn up to the lab ready to get marked. Marks will be awarded at the discretion of the demonstrator. At a minimum, one team member must be present. If no team members are available, please contact the course staff. A team evaluation task will be available on Moodle after the completion of this task.

4 Micromouse (Week 12) - 20% Total

The micromouse task involves 3 different tasks. You may compile and upload code between trials, however, no significant changes will be allowed. A significant change is one requiring more than 5 minutes to implement. Figure. 2 shows an example of what the maze may look like for the simple driving assessment. This assessment will be marked during your week 12 lab time, and is worth 20% of your final course mark.

4.1 Micromouse race - 10%

This task involves path planning through computer vision and the autonomous maze-solving competition.

4.1.1 Path Generation - 2%

Teams will take an image of the maze and use it to generate a series of commands that solve the maze. This series of commands can be copied to the robot's source code and uploaded. Teams may choose to use the command format laid out previously or a custom solution. The initial and goal locations will be provided to you in the format outlined in 1.2.

2 marks will be awarded for showing a demonstrator that your code executes. You must show the image input and the output to verify that your solution is not hard-coded. You will not be permitted to tune the values after taking an image, you must immediately execute the code and enter the output into the robot's source code in front of your demonstrator.

The generated path from this task is required for the next task, as such, if your program fails to execute or output a path, you will not be able to attempt 4.1.2.

4.1.2 Maze Completion - 8%

The path generated from 4.1.1 is entered in the robot's source code and uploaded to the robot as mentioned. The robot is then placed at the starting position and must begin navigating the maze when turned on. Marks are awarded based on maze completion and the time taken to solve the entire maze. The time is calculated from the moment the robot moves autonomously from the start to the moment it is fully enclosed in the final cell (facing the correct direction) and stationary.

Maze Completion %	Mark
[0%-19%)	0
[20%-39%)	1
[40%-59%)	2
[60%-79%)	3
[80%-99%)	4
[100%]	5

Table 1: Marks awarded for maze completion.

Speed	Mark
Top 50% of teams which completed the maze	1
Top 25% of teams which completed the maze	2
Top 10% of teams which completed the maze	3

Table 2: Marks awarded for competition speed.

To be eligible for the speed marks you must complete the maze. Marks for maze completion progress will be at the discretion of the marker. Each team will get a total of two runs. The first minor collision (hit but continues without assistance) will be allowed. The second minor collision or the first major collision (requires human intervention) will cause the run to terminate.

4.2 Continuous Planning - 5%

For this task, a 5*5 cell section of the maze will be replaced by an obstacle course that the robot must navigate (Figure. 3). All the obstacles will be cylindrical with a diameter of 100mm at the same height as the maze walls, however, they will be placed randomly within the designated area. Computer vision should be used to generate an occupancy map from the image and solve the path. You may hardcode the position of cells in which the obstacle course is. The

exact formatting used to transfer commands from the computer vision code to the robot is up to you. It may be best to generate a list of waypoints relative to each other, which the robot can move to in series. There will only ever be one entrance and exit to the obstacle course.

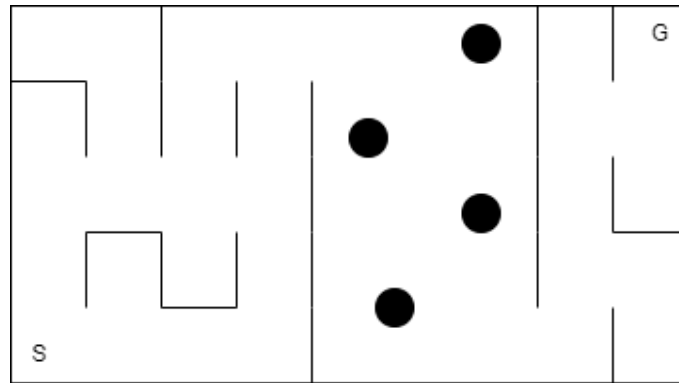


Figure 3: Example of a continuous maze your team may be presented with

1 mark will be awarded for generating an occupancy map and solving it. To receive this mark you must show an image with the trajectory to be taken by the robot to the demonstrator. 2 marks will be awarded for showing your robot can complete the obstacle course by following the trajectory shown in your occupancy map image. For this, your robot may start at the first cell in the obstacle course and finish in the last cell of the obstacle course. 2 marks will be awarded if the robot starts from the start of the maze and makes it to the end of the maze, moving through the obstacle course autonomously.

Each team will get a total of two runs. The first minor collision (hit but continues without assistance) will be allowed. The second minor collision or the first major collision (requires human intervention) will cause the run to terminate. Partial marks will be awarded at the discretion of the marker based on progress through the obstacle course if the run fails.

4.3 Autonomous Mapping - 5%

For this task, teams will use the robot to autonomously search and generate a map of the standard maze. It will then navigate to the start position, solve the maze, and complete the shortest path run of the maze. During the mapping of the maze, a visualisation of the map must be shown on the screen with a % completion score, derived from the number of cells the robot has visited.

The only information allowed to be hard-coded into your solution is the start coordinates (including orientation) and the goal coordinates. You must prove to your demonstrator that your solution is not hard-coded to get marks for this task. Marks for maze completion progress will be at the discretion of the marker. Each team will get a total of two runs. The first minor collision (hit but continues without assistance) will be allowed. The second minor collision or the first major collision (requires human intervention) will cause the run to terminate.

Maze Completion %	Mark
[0%-20%)	0
[20%-40%)	1
[40%-60%)	2
[60%-80%)	3
[80%-100%)	4

Table 3: Maze Completion Marks

1 mark will then be awarded if the robot solves the maze in the shortest path after mapping it. It is possible to explore and solve the shortest path maze without fully mapping the maze. In this case, teams will be awarded the full 5 marks.

NOTE: Attempting and succeeding in this task does not award marks for the computer vision path planning task.

4.4 Submission

Marking will occur during your Week 12 Lab. Teams will be called to the front to be marked. It is expected that you turn up to the lab ready to get marked. You may compile and upload code between trials, however, no significant

changes will be allowed. A significant change is one requiring more than 5 minutes to implement. At a minimum, one team member must be present. If no team members are available, please contact the course staff. A team evaluation task will be available on Moodle after the completion of this task.

5 Plagiarism

If you are unclear about the definition of plagiarism, please refer to [What is Plagiarism? — UNSW Current Students](#). You could get zero marks for the assignment if you were found:

- Knowingly providing your work to anyone and it was subsequently submitted (by anyone), or
- Copying or submitting any other person's work, including code from previous students of this course (except general public open-source libraries/code). Please cite the source if you refer to open-source code.

You will be notified and allowed to justify your case before such a penalty is applied.

5.1 Generative AI

Code written with a generative AI (ChatGPT) is allowed but must be clearly labelled with in-line comments and mentioned in the file header.